Mean Value Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of mean value theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AC-01 / Advanced Calculus / Report v1

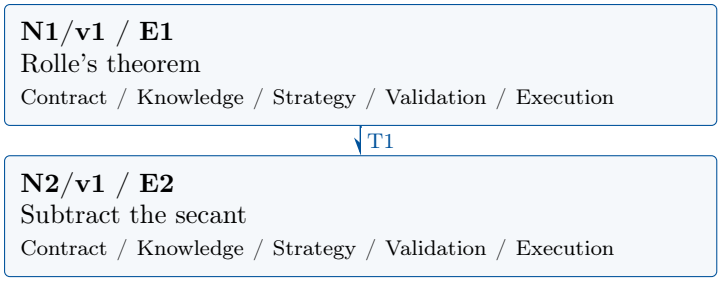
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b), $a < b$, prove there is $c \in (a, b)$ with $f'(c) = (f(b) - f(a))/(b - a)$.*

A. Foundations and permitted imports

Import the extreme value theorem for continuous real functions on a closed bounded interval. Fermat's stationary-point lemma is proved below by one-sided difference quotients. Differentiability is required only in the open interval.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. An interior point with $h'(c) = 0$.

Receiving activity. N2: Subtract the secant

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Given f and interval $a < b$; prior exports E1/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Rolle's theorem

Object and role. Find an interior extremum unless the function is constant.

Inputs. Continuous h with $h(a) = h(b)$, differentiable inside; foundational imports stated in §1

C — Contract

Own E1/v1: An interior point with $h'(c) = 0$.

K — Knowledge

Retain Continuous h with $h(a) = h(b)$, differentiable inside; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Find an interior extremum unless the function is constant.

V — Validation

A corner at an extremizer prevents the derivative conclusion.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 The extreme value theorem gives a maximum and a minimum.

N1.2 If h is constant, $h' = 0$ at every interior point.

N1.3 Otherwise at least one extremal value differs from the common endpoint value, so an extremizer c lies inside (a,b).

N1.4 At an interior local maximum, difference quotients with positive step are ≤ 0 and with negative step are ≥ 0.

N1.5 If the two-sided derivative exists it is therefore 0.

N1.6 At a minimum the inequalities reverse and give the same conclusion.

N1.7 Thus Rolle's theorem holds.

Export E1/v1. An interior point with $h'(c) = 0$.

Acceptance gate. A corner at an extremizer prevents the derivative conclusion. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: A corner at an extremizer prevents the derivative conclusion. Recheck dependent nodes N2.

B. N2/v1: Subtract the secant

Object and role. Make endpoints equal by an auxiliary affine function.

Inputs. Given f and interval $a < b$; prior exports E1/v1

C — Contract

Own E2/v1: The specified secant slope is attained by an interior derivative.

K — Knowledge

Retain Given f and interval $a < b$; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Make endpoints equal by an auxiliary affine function.

V — Validation

Division by $b - a$ is valid because $a < b$.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Set $m = (f(b) - f(a))/(b - a)$ and $h(x) = f(x) - f(a) - m(x - a)$.

N2.2 It has the required regularity and $h(a) = h(b) = 0$.

N2.3 By Rolle there is $c \in (a, b)$ with $h'(c) = f'(c) - m = 0$.

N2.4 Hence $f'(c) = m$, exactly the mean value theorem.

N2.5 No differentiability at the endpoints was needed.

Export E2/v1. The specified secant slope is attained by an interior derivative.

Acceptance gate. Division by $b - a$ is valid because $a < b$. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Division by $b - a$ is valid because $a < b$. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Rolle's theorem

1. The extreme value theorem gives a maximum and a minimum.
2. If h is constant, $h' = 0$ at every interior point.
3. Otherwise at least one extremal value differs from the common endpoint value, so an extremizer c lies inside (a,b).
4. At an interior local maximum, difference quotients with positive step are ≤ 0 and with negative step are ≥ 0.
5. If the two-sided derivative exists it is therefore 0.
6. At a minimum the inequalities reverse and give the same conclusion.
7. Thus Rolle's theorem holds.

B. §4.N2 — Subtract the secant

1. Set $m = (f(b) - f(a))/(b - a)$ and $h(x) = f(x) - f(a) - m(x - a)$.
2. It has the required regularity and $h(a) = h(b) = 0$.
3. By Rolle there is $c \in (a, b)$ with $h'(c) = f'(c) - m = 0$.
4. Hence $f'(c) = m$, exactly the mean value theorem.
5. No differentiability at the endpoints was needed.

□

V. WORKED ORCHESTRATION

For $f(x) = x^2$ on[1,3], the secant slope is $(9 - 1)/2 = 4$. The derivative 2x equals 4 at $c = 2$. The auxiliary $h = x^2 - 1 - 4(x - 1)$ has its interior minimum there.

A. Boundary and hypothesis audit

For $f(x) = |x|$ $on[-1, 1]$, the secant slope is 0 but its derivative where defined is ± 1. Validation detects the absent differentiability at 0, exactly where Rolle's extremum argument would require it.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

For $f(x) = |x|\ on[-1, 1]$, the secant slope is 0 but its derivative where defined is ± 1. Validation detects the absent differentiability at 0, exactly where Rolle's extremum argument would require it.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.